Loan Delinquency & NPA Monitor: Project Report

An Excel analysis of loan delinquency and non-performing loans (NPAs). Data: Prosper Loan Data, a public sample of US loans (2005-2014).

1. Executive summary

- The active loan book holds **57,990 loans worth $595.3M** (original loan amount).
- **3.55% of active loans (2,057) are overdue.** 320 of them are 90+ days late (NPA), which is 0.55% of the book and $2.57M.
- **Borrower rating is the strongest risk signal.** Delinquency rises from 0.71% for AA loans to 9.02% for HR loans.
- **Smaller loans are riskier:** 5.24% overdue for loans under 5K, against 1.59% for loans of 20K and above.
- **Loans that are 1-30 days late are the biggest overdue group (1,065 loans)** and the best place to focus collections.

2. Background and objective

Lenders watch overdue accounts closely, because a loan that slips from a few days late to 90+ days late is far harder to recover. This project builds an Excel dashboard that works like a collections MIS report. It aims to show:

- how many active loans are overdue and how late they are
- how much money sits in overdue and NPA loans
- which borrower groups (rating, loan size, state, year) carry the most risk
- all of the above, filterable in one click

3. Data and scope

- **Source:** Prosper Loan Data (Kaggle). 113,937 rows and 81 columns, covering loans from Nov 2005 to Mar 2014, in US dollars. This is public sample data, not company data.
- **After cleaning:** 113,066 unique loans (871 duplicates removed), with 11 of the 81 columns kept.
- **Loan groups:** Active 57,990 | Closed 38,066 | Written-off 17,010.
- **Key terms:** DPD means days past due. NPA means a loan that is 90+ days past due. Delinquency rate means overdue active loans divided by all active loans.

- **Scope:** overdue and NPA analysis covers **active loans only**. Written-off loans carry very large day counts (up to 2,704 days) that would distort the buckets, so they are analysed separately.

4. Method

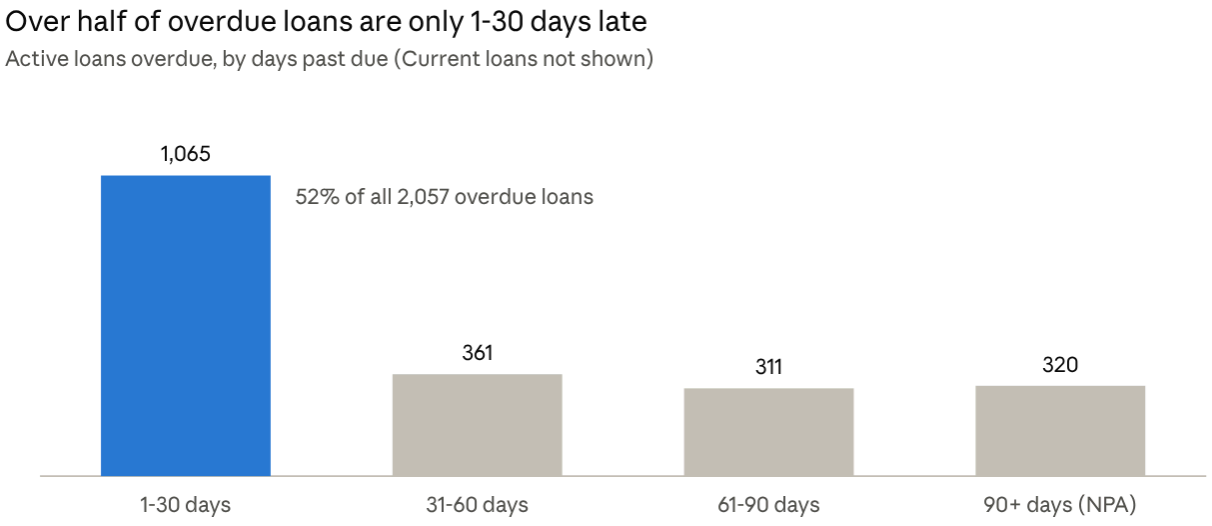
1. **Cleaning (Power Query):** removed duplicates, filled blanks ("Unknown" state, "Pre-2009" rating), and tagged each loan Active, Closed or Written-off.
2. **Helper columns (Excel formulas):** IF, IFS, AND and YEAR created DPD buckets, overdue and NPA flags, loan size bands and origination year.
3. **Analysis (12 PivotTables):** the average of a 0/1 flag gives each group's delinquency rate.
4. **Dashboard:** 5 KPI cards linked to GETPIVOTDATA cells, 4 PivotCharts and 3 slicers (rating, loan size, year).

5. Findings

5.1 Overdue loans by days past due (active book)

Bucket	Loans	% of loans	Amount	% of amount
Current	55,933	96.45%	$578.3M	97.15%
1-30 days	1,065	1.84%	$8.95M	1.50%
31-60 days	361	0.62%	$3.07M	0.52%
61-90 days	311	0.54%	$2.39M	0.40%
90+ days (NPA)	320	0.55%	$2.57M	0.43%
Total	**57,990**	**100%**	**$595.3M**	**100%**

Overdue loans hold about $17.0M, or 2.9% of active exposure. Most overdue loans are in the early bucket, which means most problems are still recoverable.



Prosper Loan Data (public sample) · active loans · 2,057 overdue loans

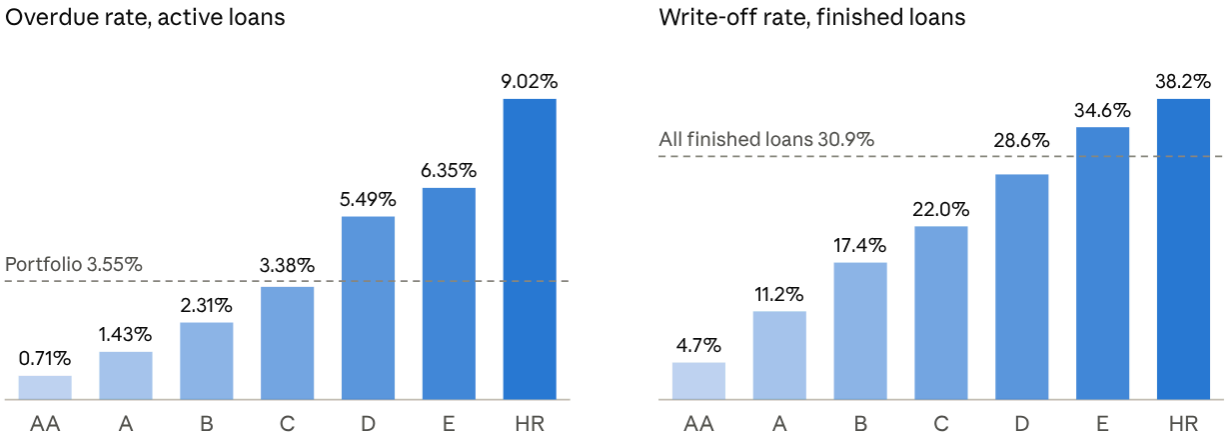
5.2 Risk by borrower rating

Rating	Active loans	Overdue rate	Write-off rate (finished loans)
AA	3,544	0.71%	4.7%
A	10,783	1.43%	11.2%
B	11,997	2.31%	17.4%
C	14,284	3.38%	22.0%
D	8,302	5.49%	28.6%
E	5,888	6.35%	34.6%
HR	3,192	9.02%	38.2%

Both measures rise with every step down in rating. HR loans are about 13 times more likely to be overdue than AA loans. The fact that today's delinquency and eventual write-offs follow the same ranking suggests that overdue rates by rating are a useful early warning.

HR loans are 13x as likely to be overdue and 8x as likely to be written off as AA

Ratings run from AA (best) to HR (worst) in both panels; darker bars mean a worse rating



Prosper Loan Data (public sample) · overdue rate on 57,990 active loans; write-off rate on 55,076 finished loans

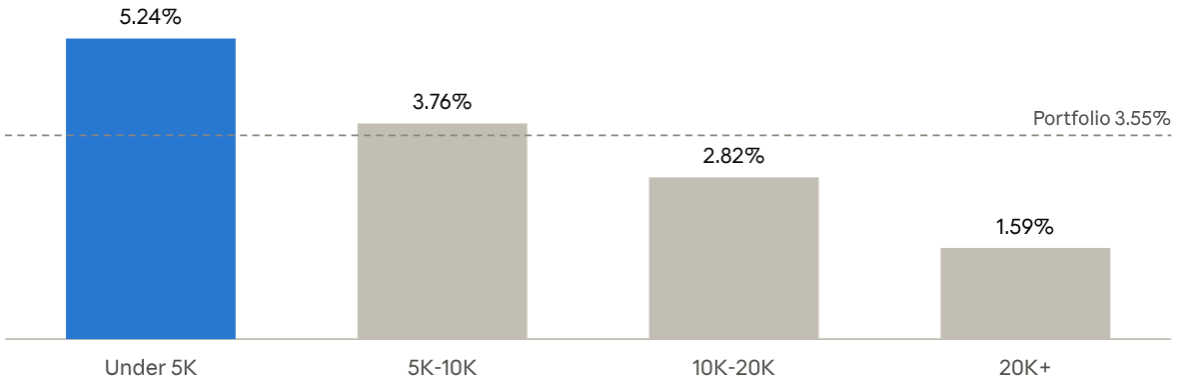
5.3 Risk by loan size

Loan size	Active loans	Overdue rate	NPA rate
Under 5K	16,263	5.24%	0.87%
5K-10K	11,343	3.76%	0.55%
10K-20K	24,043	2.82%	0.44%
20K+	6,341	1.59%	0.19%

Risk falls steadily as loan size grows. One possible reason is that borrowers taking small loans are weaker credits, but this was not tested in the data.

Loans under 5K are 3.3x as likely to be overdue as loans of 20K and above

Overdue rate of active loans, by original loan amount



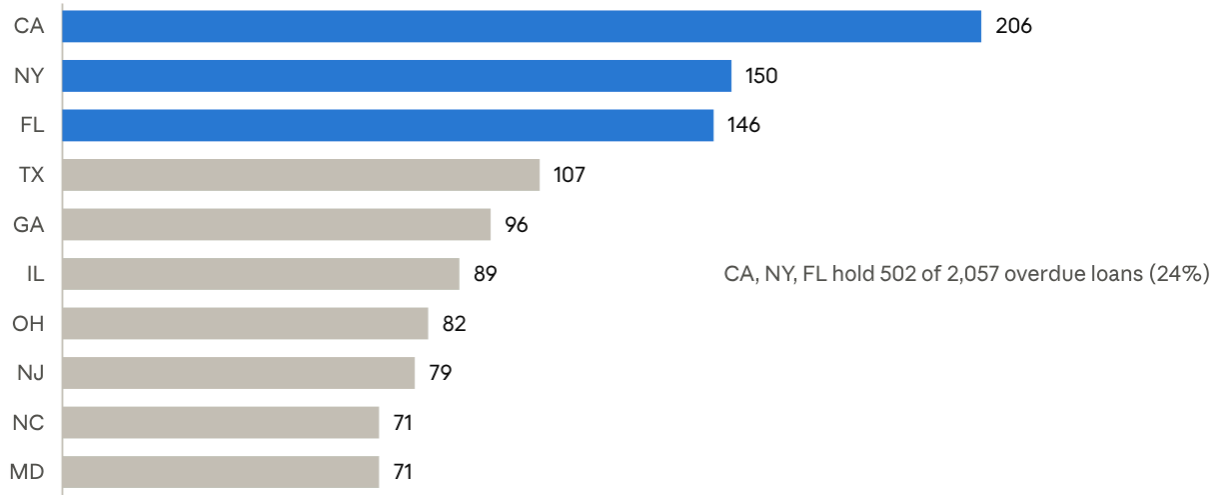
Prosper Loan Data (public sample) · 57,990 active loans · size = original loan amount

5.4 Where overdue loans are (by state)

Top 10 states by number of overdue loans: CA 206, NY 150, FL 146, TX 107, GA 96, IL 89, OH 82, NJ 79, NC 71, MD 71. These are counts, so they mostly reflect where the most loans were made, not which states are riskiest.

CA, NY and FL hold one in four overdue loans

Top 10 states by overdue active loans. Counts follow where most loans were made, not state risk.



Prosper Loan Data (public sample) · 2,057 overdue active loans · top 10 states shown

5.5 Risk by origination year

Year	Active loans	Overdue rate
2010	27	29.63%

Year	Active loans	Overdue rate
2011	3,595	8.21%
2012	11,413	7.64%
2013	31,289	2.76%
2014	11,666	0.15%

Newer loans look much safer, but this is mostly an age effect: a 2014 loan has had very little time to fall behind. 2010 has only 27 loans, which is too few to read. Years should only be compared at the same loan age.

5.6 Written-off loans

17,010 loans have been written off, which is **15.0% of all loans** and **30.9% of loans that have finished** (closed or written off).

6. Recommendations

1. **Act early on the 1-30 day bucket (1,065 loans).** It is the largest overdue group, the cheapest to recover, and stops loans rolling into later buckets.
2. **Watch the 31-60 and 61-90 day buckets (672 loans)** as the next group to become NPAs.
3. **Tighten underwriting and monitoring for D, E and HR ratings,** which have the highest delinquency and write-offs.
4. **Give small-ticket loans (under 5K) extra monitoring,** since their overdue rate is more than three times that of the largest loans.
5. **Compare loans of the same age** before judging whether newer vintages are really safer.
6. **Put regional collection effort where overdue counts are highest** (CA, NY, FL, TX, GA), then check rates before drawing conclusions about those states.

7. Limitations

- The data is a US public sample from 2005-2014, so results describe this sample and not any specific lender.
- **Exposure is the original loan amount,** not the balance still owed, because the data has no loan-level outstanding balance.
- Delinquency is a snapshot at the time the data was captured, not a trend over time.
- Collection efficiency was not calculated, as the data has no clean "amount due" figure.

- Small groups (for example NPA loans in the AA rating, or 2010 loans) can give unstable percentages. Counts should be read alongside percentages.
- The dashboard's Written-Off Rate card divides written-off loans by **all** loans (15.0%). The rate for finished loans is 30.9%. The card label should state which one is shown.

8. Conclusion

The loan book is mostly healthy: 96.45% of active loans are current and only 0.55% are NPAs. The risk is concentrated in a few places: lower-rated borrowers, small loans, and the early overdue buckets, where action is cheapest. The dashboard lets a manager see this in one view and drill into any group with slicers.

Appendix: dashboard contents

- **KPI cards:** Active Loans (57,990), Exposure ($595.29M), Delinquency Rate (3.55%), NPA Loans (320), Written-Off Rate (15%).
- **Charts:** overdue loans by DPD bucket, delinquency rate by rating, delinquency rate by loan size, top 10 states by overdue loans.
- **Slicers:** ProsperRating, Size_Band, Orig_Year.
- **Excel tools:** Power Query, Excel Tables, IF, IFS, AND, YEAR, GETPIVOTDATA, PivotTables, PivotCharts, Slicers.